

07.05.26

Applied Machine Learning

**WORKING WITH TEXT
IN HUGGING FACE**

- QUIZ
- ASSIGNMENTS
- PROJECTS
- HUGGING FACE
- OPEN DISCUSSION

QUIZ



<https://forms.office.com/e/avUHVC7Rp3>

ASSIGNMENT PRESENTATION

ASSIGNMENT RECAP

- Hugging Face Dataset Loading
- Tokenization with DistilBERT
- Dynamic Padding & Attention Masks
- Fine-tuning a Pretrained Transformer
- Hyperparameter Tuning
 - Layers to train
 - Learning rate
 - Number of epochs
- Evaluation with Accuracy and F1-score
- Loading & Saving Models Safely with safetensors

ASSIGNMENTS NEXT WEEK?

PROJECTS MILESTONES

- 23.04. Present your Ideas
- 30.04. Form Groups & Put your Name in the Excel
- 07.05. Literature Review
- 14.05. Dataset Characteristics and individual Feedback Sessions (**PUBLIC HOLIDAY**)

Deadline for completing the repo sections: 17.05.

- 21.05. Definition of Model Evaluation
- 28.05. Project Work
- 04.06. Baseline Model Estimation

Deadline for completing the repo sections: 07.06.

- 11.06. Project Work / Optional Feedback Sessions (**NO REGULAR CLASS**)
- 25.06. Project Presentations, Part I
- 02.07. Project Presentations, Part II

Submission deadline for the documented repo: 01.08.2026

BREAKOUT DISCUSSION

- Get together in groups to discuss the results of your literature review
- Quickly explain:
 - What kind of problem are you tackling in your project?
 - What kind of data are you planning to use?
 - Which models are typically used for such tasks?
 - Are there pretrained models you can use for your task?

HUGGING FACE



The AI community building the future.

The platform where the machine learning community collaborates on models, datasets, and applications.

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Tasks

[Text Generation](#)
[Any-to-Any](#)
[Image-Text-to-Text](#)
[Image-to-Text](#)
[Image-to-Image](#)
[Text-to-Image](#)
[Text-to-Video](#)
[Text-to-Speech](#)
[+ 44](#)

Parameters

[< 1B](#)
[6B](#)
[12B](#)
[32B](#)
[128B](#)
[> 500B](#)

Libraries

[PyTorch](#)
[TensorFlow](#)
[JAX](#)
[Transformers](#)
[Diffusers](#)
[Safetensors](#)
[ONNX](#)
[GGUF](#)
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Apps

[vLLM](#)
[TGI](#)
[llama.cpp](#)
[MLX LM](#)
[LM Studio](#)
[Ollama](#)
[Jan](#)
[+ 7](#)

Inference Providers

[Groq](#)
[Novita](#)
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[Nscale](#)
[fal](#)
[Hyperbolic](#)
[Together AI](#)
[moonshotai/Kimi-K2.5](#)
[Image-Text-to-Text](#) • Updated 2 days ago • [↓ 96.2k](#) • [⚡](#) • [❤ 1.42k](#)
[openai/gpt-oss-120b](#)
[Text Generation](#) • [⋮ 120B](#) • Updated Aug 26, 2025 • [↓ 2.88M](#) • [⚡](#) • [❤](#)
[Lightricks/LTX-2](#)
[Image-to-Video](#) • Updated 14 days ago • [↓ 2.72M](#) • [⚡](#) • [❤ 1.41k](#)
[microsoft/VoiceVoice-ASR](#)
[Automatic Speech Recognition](#) • [⋮ 9B](#) • Updated 6 days ago • [↓ 175k](#) • [🔄](#)
[nvidia/personalex-7b-v1](#)
[Audio-to-Audio](#) • Updated 4 days ago • [↓ 101k](#) • [❤ 1.58k](#)
[MiniMaxAI/MiniMax-M2.1](#)
[Text Generation](#) • Updated 5 days ago • [↓ 84k](#) • [⚡](#) • [❤ 1.24k](#)
[Qwen/Qwen3-ASR-0.6B](#)
[Automatic Speech Recognition](#) • [⋮ 0.9B](#) • Updated 3 days • [↓ 11.9k](#) • [❤ 128](#)
[deepseek-ai/DeepSeek-OCR-2](#)
[⋮ 3B](#) • Updated 4 days ago • [↓ 18.3k](#) • [❤ 621](#)
[arcee-ai/Trinity-Large-Preview](#)
[Text Generation](#) • [⋮ 399B](#) • Updated 5 days ago • [↓ 317](#) • [❤ 115](#)
[stepfun-ai/Step-3.5-Flash](#)
[⋮ 199B](#) • Updated about 1 hour ago • [↓ 44](#) • [❤ 114](#)

HOW TO FIND GOOD EMBEDDING MODELS

Spaces

mteb/leaderboard

like 7.35k

Running on CPU UPGRADE

AppFilesCommunity 182

Hugging Face is way more fun with friends and colleagues! Join an organizationDismiss this message

General Purpose

MultilingualEnglishHuman BenchmarkImageAudioDomain-SpecificLanguage-specificMiscellaneous

Retrieval

RTEB MultilingualRTEB EnglishImageViDoRe V3Jina Visual Document RetrievalOtherDomain-SpecificLanguage-specific

Embedding Leaderboard

This leaderboard compares 100+ text and image embedding models across 1000+ languages. We refer to the publication of each selectable benchmark for details on metrics, languages, tasks, and task types. Anyone is welcome to add a model, add benchmarks, help us improve zero-shot annotations or propose other changes to the leaderboard.

MTEB(Multilingual, v2)

The large-scale multilingual expansion of MTEB known as MMTEB, driven mainly by highly-curated community contributions covering 250+ languages.

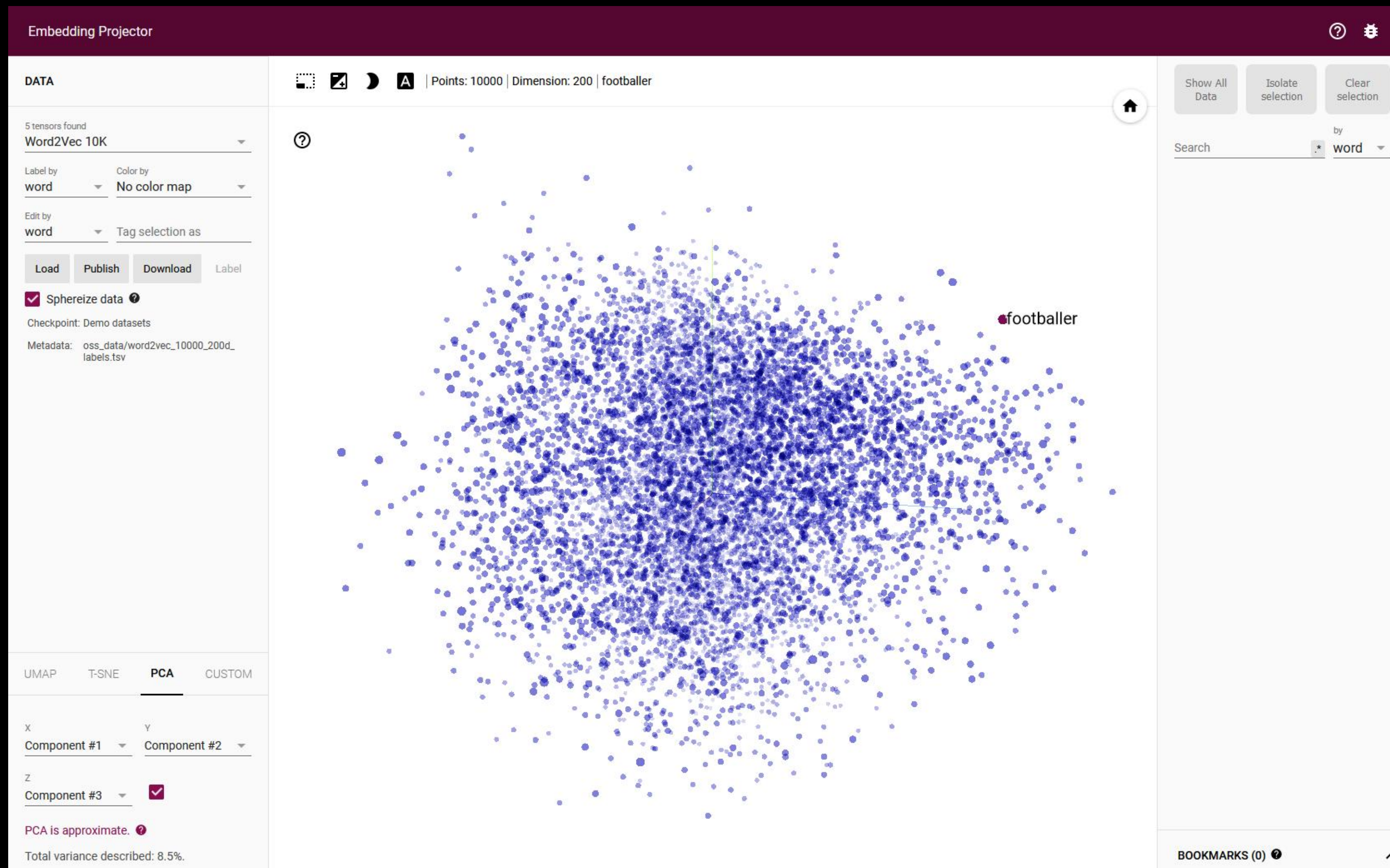
- Number of languages: 1038
- Number of tasks: 131
- Number of task types: 9
- Number of domains: 20

Click for More Info

SummaryPerformance per Model SizePerformance over TimePerformance per Task TypePerformance per taskPerformance per languageTask information

Rank (Boxed)	Model	Zero-shot	Active Param.	Total Param.	Embedding Dim.	Max Tokens	Mean (TaskType)	Mean (TaskType)	Bitext ...	Classification	Clustering	Instru...
1	harrier-oss-v1-27b	78%	25.6	27.0	5376	131072	74.27	64.20	86.02	79.95	58.93	5.39
2	KaLM-Embedding-Gemma3-12B-2511	73%	10.8	11.8	3840	32768	72.32	62.51	83.76	77.88	55.77	5.49
3	llama-embed-nemotron-8b	99%	7.0	7.5	4096	32768	69.46	61.09	81.72	73.21	54.35	10.82
4	Qwen3-Embedding-8B	99%	6.9	7.6	4096	32768	70.58	61.69	80.89	74.00	57.65	10.06
5	gemini-embedding-001	99%			3072	2048	68.37	59.59	79.28	71.82	54.59	5.18
6	Qwen3-Embedding-4B	99%	3.6	4.0	2560	32768	69.45	60.86	79.36	72.33	57.15	11.56
7	Octen-Embedding-8B	99%	6.9	7.6	4096	32768	67.84	60.28	80.35	66.68	55.68	8.90
8	F2LLM-v2-14B	88%	13.2	14.0	5120	40960	68.74	59.45	77.15	73.00	60.91	0.62
9	F2LLM-v2-8B	88%	6.9	7.6	4096	40960	68.09	58.99	75.96	71.93	60.62	0.85
10	harrier-oss-v1-0.6b	78%	0.440	0.596	1024	32768	69.01	59.00	82.85	73.88	54.00	0.81

EMBEDDING PROJECTOR



QUESTIONS ON EMBEDDINGS?

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OPEN DISCUSSION

- Would it make sense to use Transfer Learning for the Embedding layer? What would it mean to do that?
- Here are two good open discussion questions:
- When fine-tuning a pretrained model like DistilBERT, what are the trade-offs between freezing most layers and training only the classification head versus unfreezing more transformer layers?
- Why might dynamic padding be more efficient than padding every text sample to the same fixed maximum length, and are there situations where fixed padding might still be useful?

TASKS UNTIL NEXT WEEK

- Prepare questions for your projects
- Already get started on the new content